

Functional Dependency Notes

What is a Functional Dependency?

FORMAL DEFINITION

A Functional Dependency (FD) exists when the value of one attribute (or set of attributes) uniquely determines the value of another attribute (or set of attributes).

Whenever two rows have the same value of X , they must have the same value of Y .

Dependency:

$\text{RollNo} \rightarrow \text{Name}$

If RollNo is 006, Name must always be Aadya.

Candidate Keys using Functional Dependencies

DEFINITION

A Candidate Key is the smallest set of attributes that can determine all attributes in a relation. We use Functional Dependencies to find candidate keys.

Relation: $\text{RESULT}(\text{StudentID}, \text{CourseID}, \text{Grade})$

FD: $(\text{StudentID}, \text{CourseID}) \rightarrow \text{Grade}$

Neither StudentID nor CourseID alone determines Grade.

Therefore,

Candidate Key: $(\text{StudentID}, \text{CourseID})$

RULE

A Candidate Key is:

- Unique (identifies every row).
- Minimal (no unnecessary attributes).

Attribute Closure

DEFINITION

The closure of an attribute set X , denoted by X^+ , is the set of all attributes that can be functionally determined from X using the given Functional Dependencies.

Relation: $R(A, B, C)$

FDs:

- $A \rightarrow B$
- $B \rightarrow C$

Find: A^+

Solution

Start: $A^+ = \{A\}$

Apply: $A \rightarrow B$

Now: $\{A, B\}$

Apply: $B \rightarrow C$

Final: $A^+ = \{A, B, C\}$

TO COMPUTE X^+ :

1. Start with X itself.
2. Look for any FD whose left side is already in the closure.
3. Add the right-side attributes.
4. Repeat until no new attributes can be added.

Types of Functional Dependencies

1. Trivial Functional Dependency

DEFINITION

A Functional Dependency is trivial if the right-hand side is a subset of the left-hand side.

Mathematically: If $Y \subseteq X$, then $X \rightarrow Y$ is trivial.

Example

`(StudentID, Name) → Name`

Since Name is already part of the left-hand side, the dependency is trivial.

2. Non-Trivial Functional Dependency

DEFINITION

A Functional Dependency is non-trivial if the right-hand side is not a subset of the left-hand side.

Example

`StudentID → Name`

Name is not part of StudentID, so this is non-trivial.

3. Completely Non-Trivial Functional Dependency

DEFINITION

A Functional Dependency is completely non-trivial if the left-hand side and right-hand side have no common attributes.

$$X \cap Y = \emptyset$$

Armstrong's Axioms

1. Reflexivity

RULE

If: $Y \subseteq X$

Then: $X \rightarrow Y$.

2. Augmentation

RULE

If: $X \rightarrow Y$

Then for any attribute set Z : $XZ \rightarrow YZ$.

Example

Given: `StudentID → Name`

Add CourseID to both sides:

`(StudentID, CourseID) → (Name, CourseID)`.

3. Transitivity

RULE

If: $X \rightarrow Y$ and $Y \rightarrow Z$

Then: $X \rightarrow Z$.

Derived Rules

These are obtained from Armstrong's Axioms.

UNION (ADDITIVITY)

If: $X \rightarrow Y$ and $X \rightarrow Z$

Then: $X \rightarrow YZ$.

DECOMPOSITION (PROJECTIVITY)

If: $X \rightarrow YZ$

Then: $X \rightarrow Y$ and $X \rightarrow Z$.

COMPOSITION

If: $X \rightarrow Y$ and $A \rightarrow B$

Then: $XA \rightarrow YB$

Example

$\text{StudentID} \rightarrow \text{Name}$

$\text{CourseID} \rightarrow \text{CourseName}$

Infer: $(\text{StudentID}, \text{CourseID}) \rightarrow (\text{Name}, \text{CourseName})$.

PSEUDOTRANSITIVITY

Rule

If: $X \rightarrow Y$ and $WY \rightarrow Z$

Then: $WX \rightarrow Z$.

Example

Given:

$\text{StudentID} \rightarrow \text{Department}$

$(\text{Department}, \text{Semester}) \rightarrow \text{Classroom}$

Infer: $(\text{StudentID}, \text{Semester}) \rightarrow \text{Classroom}$